

ROSMASX3 Project Summary

1. Current System Overview

The system currently consists of three main ROS2 nodes:

- Hallway Detector: Uses OpenCV edge detection to determine free space and outputs a steering direction and bias.
- Hallway Follower: Uses PID control combined with LiDAR data to follow hallways and avoid obstacles.
- Scan Monitor: Confirms LiDAR, IMU, and odometry data are being received.
- Might add a fourth node to replace the camera. The node would use NVIDIA's **Cosmos Reason-2-8B**, testing the code to see if the PID, works by itself first without the camera, then replacing it.

The system performs basic autonomous hallway navigation using a combination of vision and LiDAR.

2. Problems Encountered

Several key issues have been encountered:

- Camera instability: Noisy edge detection leads to inconsistent steering decisions.
- PID tuning challenges: Oscillation and overcorrection due to noisy inputs.
- Sensor fusion conflicts: Camera and LiDAR sometimes provide conflicting guidance.
- ROS2 integration issues: Launch errors and topic mismatches during setup.
- Lack of true perception: Current system is rule-based and does not understand the environment.

3. Final Goal

The goal is to build a fully autonomous indoor robot capable of:

- Navigating hallways smoothly
- Avoiding obstacles reliably
- Using an AI model for perception instead of heuristic vision
- Integrating with SLAM for mapping and localization

Future architecture will replace the current vision pipeline with a neural network and expand into full autonomous navigation.

4. Next Steps

- Isolate and tune PID control without camera input
- Stabilize sensor inputs
- Replace hallway detector with an AI model
- Integrate SLAM for mapping and navigation